

Homework #1

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Submission by (netIDs): **FILL YOUR NETID HERE**

Logistics: Read all the instructions carefully before you start working on the assignment, and before you make a submission.

- The assignment is due on February 21, 11.59 p.m. **Please read the submission instructions included at the end of this document carefully.**
- You can do this homework either individually or in a group of 2. We strongly recommend doing this in a group. All students in a group will receive the same grade. You will **declare your group by February 16, 11.59 p.m.** on gradescope.
- This assignment is worth 10% of your grade. Make sure to submit this assignment by the due date mentioned above. Remember that you have 5 slip days to use throughout the course. You may use a maximum of 2 slip days for each assignment. Refer to the course webpage for more information about late submission policy.
- Please typeset your assignment in L^AT_EX. You can use this template for your written answers. Please include the netID(s) of all members in the group at the top of this page. (Search for "FILL YOUR NETID" to find the appropriate place to enter your netIDs).
- You will submit this assignment on gradescope. We have created 3 assignments: "hw1-partnerdeclaration", "hw1-written" and "hw1-programming". You will submit the written parts of the assignment in the form of a pdf to "hw1-written". This will include your answers to Section A and the written questions in Section B. You will submit code or outputs of code to "hw1-programming". Please read the submission instructions included at the end of this document for details on this.
- The University Academic Code of Conduct will be strictly enforced.

Section A: Conceptual Questions

(40 points)

N-gram Language Models

(1) Consider the following small corpus of sentences:

$\langle s \rangle$ the cat sat on the mat $\langle /s \rangle$
 $\langle s \rangle$ the dog sat $\langle /s \rangle$
 $\langle s \rangle$ the mat is on the floor $\langle /s \rangle$

where $\langle s \rangle$ and $\langle /s \rangle$ are tokens representing the start and end of a sentence respectively. Remember that in all probability computations of n-gram models, we will ignore the unigram probability of the token $\langle s \rangle$.

(a) (4 points) Provide all **non-zero** bigram probabilities, i.e. $P(w_{i+1}|w_i)$ for this corpus. Your vocabulary set for this corpus is $\mathcal{V} = \{\text{the, cat, dog, sat, on, is, mat, floor, } \langle s \rangle, \langle /s \rangle\}$.

(Solution) $P(\text{the} | \langle s \rangle) = 1$

$P(\text{cat} | \text{the}) = \frac{1}{5}$

$P(\text{sat} | \text{cat}) = 1$

$P(\text{on} | \text{sat}) = \frac{1}{2}$

$P(\text{the} | \text{on}) = 1$

$$\begin{aligned}
 P(mat|the) &= \frac{2}{5} \\
 P(</s>|mat) &= \frac{1}{2} \\
 P(dog|the) &= \frac{1}{5} \\
 P(sat|dog) &= 1 \\
 P(</s>|sat) &= \frac{1}{2} \\
 P(is|mat) &= \frac{1}{2} \\
 P(on|is) &= 1 \\
 P(floor|the) &= \frac{1}{5} \\
 P(</s>|floor) &= 1
 \end{aligned}$$

(b) (2 points) Give 2 examples of grammatically valid sentences in the English language that will be assigned a zero probability according to this bi-gram language model. Your examples should only use tokens in the above vocabulary set $\mathcal{V}$.

(Solution) $<s>$ the dog is on the floor $</s>$ and $<s>$ the cat is on the floor $</s>$

(c) (4 points) The sentences seen in the training corpus (i.e. the 3 sentences above) will always have a strictly higher probability than any unseen grammatical sentence according to the above 2-gram model. True or False? Justify your choice with either a proof (if true) or a counterexample (if false).

Counterexample: $<s>$ the cat sat $</s>$

$$\text{Prob: } P(the|<s>)P(cat|the)P(sat|cat)P(</s>|sat) = (1)(\frac{1}{5})(1)(\frac{1}{2}) = \frac{1}{10}$$

compared with $<s>$ the dog sat $</s>$

$$\text{Prob: } P(the|<s>)P(dog|the)P(sat|dog)P(</s>|sat) = (1)(\frac{1}{5})(1)(\frac{1}{2}) = \frac{1}{10}, \text{ therefore not strictly greater.}$$

(d) (6 points) Suppose you are given a prefix $w_0w_1w_2w_3w_4w_5 = "<s> the mat is on the"$. We can use the bigram model to generate possible sentences, i.e. continuations ending in $</s>$, for this prefix. The pseudocode for this algorithm is:

```

1. Initialize str =  $w_0...w_4w_5$ 
2.  $i = |\text{str}| - 1$       #  $i = 5$  for our example
3. while ( $w_i \neq </s>$ )
4.     Sample  $w_{i+1} \sim P(w_{i+1}|w_0...w_i)$ 
5.     str = str +  $w_{i+1}$ 
6.      $i = i + 1$ 
7. return str

```

Essentially, in each iteration of the while loop, we sample one word according to your bigram's next word probability. We continue to do this till you encounter the $</s>$ token. This gives us one possible sentence according to the bigram language model.

Write 2 different sentences (sequences ending in $</s>$) that have a non-zero probability according to your bigram language model starting from the above prefix ($w_0w_1w_2w_3w_4w_5 = "<s> the mat is on the"$). Report both the sentences and the probability of the continuation, i.e., report $P(w_6...w_N | <s> the mat is on the)$, where $w_N = </s>$.

(Solution) Sentence 1: $<s>$ the mat is on the mat $</s>$

$$\text{Prob 1: } P(mat | <s> | <s> the mat is on the) = P(mat|the)P(</s>|mat) = (\frac{2}{5})(\frac{1}{2}) = \frac{1}{5}$$

Sentence 2: $<s>$ the mat is on the floor $</s>$

$$\text{Prob 1: } P(floor | <s> | <s> the mat is on the) = P(floor|the)P(</s>|floor) = (\frac{1}{5})(1) = \frac{1}{5}$$

(e) (3 points) For the prefix in part (d), what is the total number of distinct non-zero probability sentences that we can generate? Comment on the grammaticality of these, would a higher order n-gram model help or hurt?

(Solution) Infinite because we can keep looping: the mat is on the mat is on the... etc. This would be grammatically incorrect. A higher order n-gram would help. For example a 4-gram model would predict "floor" after " $<s>$ the mat is on the" because given "is on the" could only produce "floor" (and the next iteration would end the sentence).

Text Classification

(2) In this question we will consider the task of determining if a given sentence is imperative, i.e. is a command (label = 1) or not (label = 0). For example, the sentence “*Alexa, turn off the lights.*” is imperative, but the sentence “*The lights are turned off.*” is not. We want to perform binary logistic regression to complete this task. Assume we have selected the feature vector $\langle f_0, f_1, f_2, f_3, f_4 \rangle$, with the features defined below:

Feature w	Description c	Numerical Value
f_0	infinitive verb at beginning of sentence	1 if true, else 0
f_1	subject absent	1 if true, else 0
f_2	object after verb	1 if true, else 0
f_3	sentence length	# of words
f_4	ends with a ! or .	1 if true, else 0

(a) (4 points) Consider the sentence “Please submit your homework by Friday, February 20th”. With the above model, the extracted feature vector is $\langle 0, 1, 1, 8, 1 \rangle$. Compute the probability that our binary logistic regression model will classify this sentence as imperative, given the weight vector $\langle 0.9, 1.1, 0.6, -0.25, 0.2 \rangle$. With a threshold of 0.5, what does the model classify the sentence as? Show all work.

(Solution) First compute the dot product between the weight vector and the feature vector:

$$\begin{aligned}\mathbf{w}^\top \mathbf{f} &= (0.9)(0) + (1.1)(1) + (0.6)(1) + (-0.25)(8) + (0.2)(1) \\ &= -0.1.\end{aligned}$$

To compute the probability of $P(y = 1 \mid \cdot)$, we use the sigmoid function:

$$P(y = 1 \mid \mathbf{f}) = \frac{1}{1 + e^{-(-0.1)}} = \frac{1}{1 + e^{0.1}} \approx \frac{1}{2.105} \approx 0.475.$$

Since $0.475 < 0.5$, the model classifies the sentence to have label $y = 0$.

(b) (5 points) Design a feature so as to augment the current feature set with an intuitively “informative” feature. Derive the new feature vector $\langle 0, 1, 1, 8, 1, x \rangle$ for the above sentence in part (a). Recompute the probability you found in part (a) with the additional feature now assigned a weight 1.0. If needed, what changes would you make to your additional feature to ensure that this classification is correct.

(Solution) Many options here. For example, you can design a binary feature that takes value 1 if the input sentence starts with the word “Please”, and 0 otherwise. You can re-do a similar calculation as above – you’d find that the new logistic regression model will classify it as imperative ($y = 1$). Note that we expect your feature to be **intuitively** informative. For example, simply having a constant feature that is always 1 is not an acceptable answer.

(c) (4 points) You decide to train your model using stochastic gradient descent. You augment the feature vector with a new feature f_5 = the sentence contains “Please”. After training on many examples and over many epochs, your learned weights are $\langle -0.3, 2.8, 1.9, -0.18, 0.4, 0.08 \rangle$. You find it odd that the “Please” weight is so low, so you add another training example:

sentence w	feature vector	true label	learning rate
“Please close the window later.”	$\langle 0, 1, 1, 5, 1, 1 \rangle$	1 (is imperative)	$\alpha = 0.1$

Table 1: Training Data

Compute the gradient for the weight associated with f_5 after one step of training on the example in Table 1 and using the learning rate $\alpha = 0.1$. Recall that for logistic regression:

$$\frac{\partial L_j}{\partial w_i} = f_i^j \left[\sigma \left(\sum_k w_k f_k^j \right) - y_j \right]$$

(Solution)

First, we need to compute the probability of $P(y = 1|x)$. If we redo the above computation, this will compute to ≈ 0.986 .

Now, note that the gradient formula is basically the following when the ground truth label y_j is 1:

$$\frac{\partial L_j}{\partial w_i} = f_i^j (P(y = 1|x) - 1) = -0.014$$

The above is the gradient.

(d) (2 point) What could be one explanation for why the weight for the contains “Please” feature remained close to 0 even with many examples? Provide an example that supports your reasoning.

(Solution) Most reasonable explanations will be accepted here. You basically need to demonstrate that there is a training dataset for which “please” does not provide any signal on how to distinguish between the two classes.

Examples – in the training data, there are a similar number of sentences belonging to both classes (0 / 1) includes “please”, so this wasn’t a distinguishing feature. Or that very few sentences in the training data included “please”.

Word Embeddings

(3) You are given the following data (Table 2) to train a **skip-gram model** with negative sampling. Let C and W be the context and word embedding matrices respectively, where row c_t and w_t denote the context and word vector for token t .

Target word w	Context Word c	label y
ball	beach	+
ball	dog	+
cafe	cat	+
park	cafe	-
ball	cat	-
cat	dog	+
cat	ball	-

Table 2: Training Data

(a) (4 points) After training on multiple epochs on the above training data, which row vectors in C and W will have changed the most from their initial values? Explain your reasoning

(Solution) There is a typo here, we meant to ask “changed from their initial values” not “changed most”.

If the question was ”changed from their initial values”, the answer would have been $w_{\text{ball}}, w_{\text{cafe}}, w_{\text{park}}, w_{\text{cat}}, c_{\text{dog}}, c_{\text{beach}}, c_{\text{cat}}, c_{\text{cafe}}, c_{\text{ball}}$.

The question, as stated, “changed most” is not answerable with the provided information.

Due to this error on our part, we will accept all solutions that display understanding of the fact that only vectors in the above list *can* change. More specifically, if your answer includes a vector not in the above list, e.g. c_{park} , we will deduct points.

(b) (2 points) In the training data, notice that ‘cat’ appears as a context word in multiple examples with different labels. What is the model expected to learn about the relationships between these words?

(Solution)

The training will increase the similarity between w_{cafe} and c_{cat} and decrease the similarity between w_{ball} . Therefore, w_{cafe} and w_{ball} will likely be dissimilar to each other.

You can extrapolate from this and reason about other pairs as well.

Section B: Programming

(100 points)

You will essentially follow the `hw1_studentfacing_sp26_cs4740.ipynb` file in this [google drive folder](#) to complete this coding assignment. Please read the description below to understand the task setup.

In this programming assignment, you will implement Binary Logistic Regression from scratch for the natural language entailment task. Given a sentence pair (P, H) where P is a premise and H is a hypothesis, we are aiming to classify the relation between them. For this assignment, we will only consider two labels, **entailment**, meaning the hypothesis must be true if the premise is true, and **contradiction**, meaning the hypothesis cannot be true if the premise is true.

(Solution) [Write the written solutions to the programming assignment here.](#)

Submission

We will create 3 different Gradescope assignments for this homework. Below we outline submission details for each.

2. Assignment hw1-partnerdeclaration

Due date: February 16, 2026. 11.59 p.m.

Ungraded but mandatory!! Please declare if you plan to submit this assignment with a partner or by yourself. If former, declare your partner's NetID.

2. Assignment hw1-written

Due date: February 21, 2026. 11.59 p.m.

Max points: 40

This assignment is for you to submit the written parts of this homework. This includes: (1) Solutions to Section A, (2) Solutions to the written questions in Section B. Specifically, this means questions Q1 (Q1.1 - Q1.2), Q2 (Q2.1), Q3 (Q3.1, Q3.2, Q3.3 and Q3.4) and Q4 (Q4.1).

3. Assignment hw1-programming

Due date: February 21, 2026. 11.59 p.m.

Max points: 100

You will follow the instructions in the `hw1_studentfacing_sp25_cs4740.ipynb` in the google drive. It will ask you to implement functions in 2 different python files (also included in the google drive): *features.py* and *models.py*. You will submit these 2 Python files to **hw1-programming**.

Please double check to make sure that you have implemented all the functions that need to be implemented in these files.